

- 2) Derive the Maxwell equation obtained from the principle of electrostatic and magnetostatic!

3) Gauss law: $\oint_S \vec{D} \cdot d\vec{s} = Q_{\text{encl}} \rightarrow (1)$

$$\therefore Q = \int_V \rho_v dv$$

We know that Surface relate to Volume

$$\oint_S \vec{D} \cdot d\vec{s} = \int_V \rho_v dv$$

Applying divergence theorem

$$\oint_S \vec{D} \cdot d\vec{s} = \int_V (\nabla \cdot \vec{D}) dv$$

$$\therefore \int_V (\nabla \cdot \vec{D}) dv = \int_V \rho_v dv$$

$$\int_V (\nabla \cdot \vec{D}) dv = \int_V \rho_v dv$$

$$\boxed{\nabla \cdot \vec{D} = \rho_v}$$

2) Magnetic flux: $\oint_S \vec{B} \cdot d\vec{s} = 0 \rightarrow (2)$

Applying Divergence theorem

$$\oint_S \vec{B} \cdot d\vec{s} = \int_V (\nabla \cdot \vec{B}) dv$$

$$\int_V (\nabla \cdot \vec{B}) dv = 0$$

$$\boxed{\nabla \cdot \vec{B} = 0}$$

3) Electromagnetic field:

$$\oint_C \vec{E} \cdot d\vec{l} = -\frac{d}{dt} \int_S \vec{B} \cdot d\vec{s}$$

Apply Stoke's theorem:

$$\oint_C \vec{E} \cdot d\vec{l} = \int_S (\nabla \times \vec{E}) \cdot d\vec{s}$$

$$\int_S (\nabla \times \vec{E}) \cdot d\vec{s} = -\int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{s}$$

$$\boxed{\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}}$$

4) Ampere maxwell law : $\oint \vec{H} \cdot d\vec{l} = I_{enc}$

Where $I_d = \frac{d}{dt} \int_S \vec{D} \cdot d\vec{S}$

$$\oint \vec{H} \cdot d\vec{l} = \int_S \vec{J} \cdot d\vec{S} + \frac{\partial}{\partial t} \int_S \vec{D} \cdot d\vec{S}$$

Apply Stokes theorem

$$\int_S (\nabla \times \vec{H}) \cdot d\vec{S} = \int_S \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right) \cdot d\vec{S}$$

$$\boxed{\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}}$$

2) Identify various components contributing force in static magnetic field and hence obtain Lorentz force equation!

Sol:

1) Electric field force $\vec{F} = q\vec{E}$

2) Magnetic field force $\vec{F} = q(\vec{v} \times \vec{B})$

Derivation of Lorentz force equation

$$\vec{F} = \vec{F}_E + \vec{F}_B$$

$$= q\vec{E} + q(\vec{v} \times \vec{B})$$

$$\boxed{\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})}$$

3) Use Biot-Savart's law to calculate the magnetic field intensity at specified point due to current carrying conductor!

We know that formula $d\vec{H} = \frac{I d\vec{l} \times \vec{a}_r}{4\pi R^2}$

Apply Biot-Savart's law $d\vec{l} = dz \hat{a}_z$

$$|d\vec{H}| = \frac{I d\vec{l} \sin 90^\circ}{4\pi R^2}$$

$$[\angle d\vec{l} \text{ \& } \angle a_r = 90^\circ]$$

$$dH = \frac{I dz}{4\pi R^2}$$

$$\text{distance } R = \sqrt{r^2 + z^2}$$

$$dH = \frac{I dt}{4\pi(r^2 + z^2)}$$

Integrate on RHS

$$H = \int_{-\infty}^{\infty} \frac{I dt}{4\pi(r^2 + z^2)} = \frac{I}{4\pi} \int_{-\infty}^{\infty} \frac{dz}{r^2 + z^2} = \frac{\pi}{r} \frac{I}{4\pi} = \frac{I}{2\pi r}$$

$$\boxed{\vec{H} = \frac{I}{2\pi r} \phi \text{ A/m}}$$

4) Derive Ampere's Circuital law to determine the net current flow through current element.
sol: Magnetic field around any closed path is equal to current enclosed

$$\oint \vec{H} \cdot d\vec{u} = I_{\text{enc}} \rightarrow \text{①}$$

$$[\because \vec{H} \cdot d\vec{u} = H \cdot dl]$$

$$\oint \vec{H} \cdot d\vec{u} = \oint H \cdot dl = H \oint dl = H(2\pi r)$$

$$\oint \vec{H} \cdot d\vec{u} = \frac{I}{2\pi r} (2\pi r)$$

$$\boxed{\oint \vec{H} \cdot d\vec{u} = I_{\text{enc}}}$$

5) Apply electrostatic and magnetostatic principle to compute boundary conditions with reference E , D , H , and B any surface discontinuity?

Electrostatic boundary

1) We know that $\oint \vec{E} \cdot d\vec{u} = 0$

Result $\vec{E}_{1t} = \vec{E}_{2t} \rightarrow \text{tangential}$

2) Boundary Condition Electric flux density

$$\oint \vec{D} \cdot d\vec{s} = Q_{\text{enc}}$$

$$D_{1n} - D_{2n} = \rho_s$$

$$\boxed{D_{1n} = D_{2n}}$$

Magnetostatic

$$1) \oint \vec{H} \cdot d\vec{a} = I_{encl}$$

$$\vec{H}_{2t} - \vec{H}_{1t} = \vec{k}_s \times \vec{a}_n$$

[$\therefore$ no surface current]

$$\boxed{\vec{H}_{1t} = \vec{H}_{2t}}$$

$$2) \oint \vec{B} \cdot d\vec{s} = 0$$

$$\boxed{B_{1n} = B_{2n}}$$